

HW#10**REF:** *Theory Manual Chapter 2: §2.10 – §2.13**Appended 2011 Gang Rod Integral Worth Curve and Xenon Worth Curves**PULSTAR Reactivity Coefficients PDF (posted on Moodle site)*

1. **Calculate the ECP** (estimated critical position) Gang Rod Height (inches from bottom) for Tuesday morning at 8:00 AM given the following reactor operating history and conditions (*Use the attached 2011 Gang Integral Rod Worth curve and Xenon reactivity worth curve, and reactivity coefficients as given in class*):

Monday:Prior to startup:

Pool Temp = 95°F
Xenon = 0 pcm
Experiments = -100 pcm
Source = 0 pcm (inserted)

Tuesday:Prior to startup @ 8:00 AM Tuesday:

Pool Temp = 102°F
Xenon = ?? pcm
Experiments = +50 pcm
Source = -16 pcm (removed)

- **ACP (Actual Critical Position) = 18.5"**
Rods Ganged @ 100 Watts
- **Operated 8 hrs @ 500kW**
- **Shutdown Monday @ 4:00 PM**

Calculate ECP (gang rod height) @ 100 Watts for Tues. @ 8:00 AM.

2. **Calculate Excess Reactivity (ρ_{excess})** for Tuesday morning @ 8:00 AM correcting to Cold, Clean, Critical conditions:

DEF: Cold, Clean, Critical Conditions

- No Xenon in core ($\rho_{\text{xenon}} = 0$ pcm)
- No Experiments in core ($\rho_{\text{exper}} = 0$ pcm)
- Pool (Moderator) Temp (T_2) = 70°F
- Startup Source In ($\rho_{\text{source}} = 0$ pcm)

Hint: Where at Cold, Clean, Critical, (ρ_{excess}) = - (ρ_{rods})

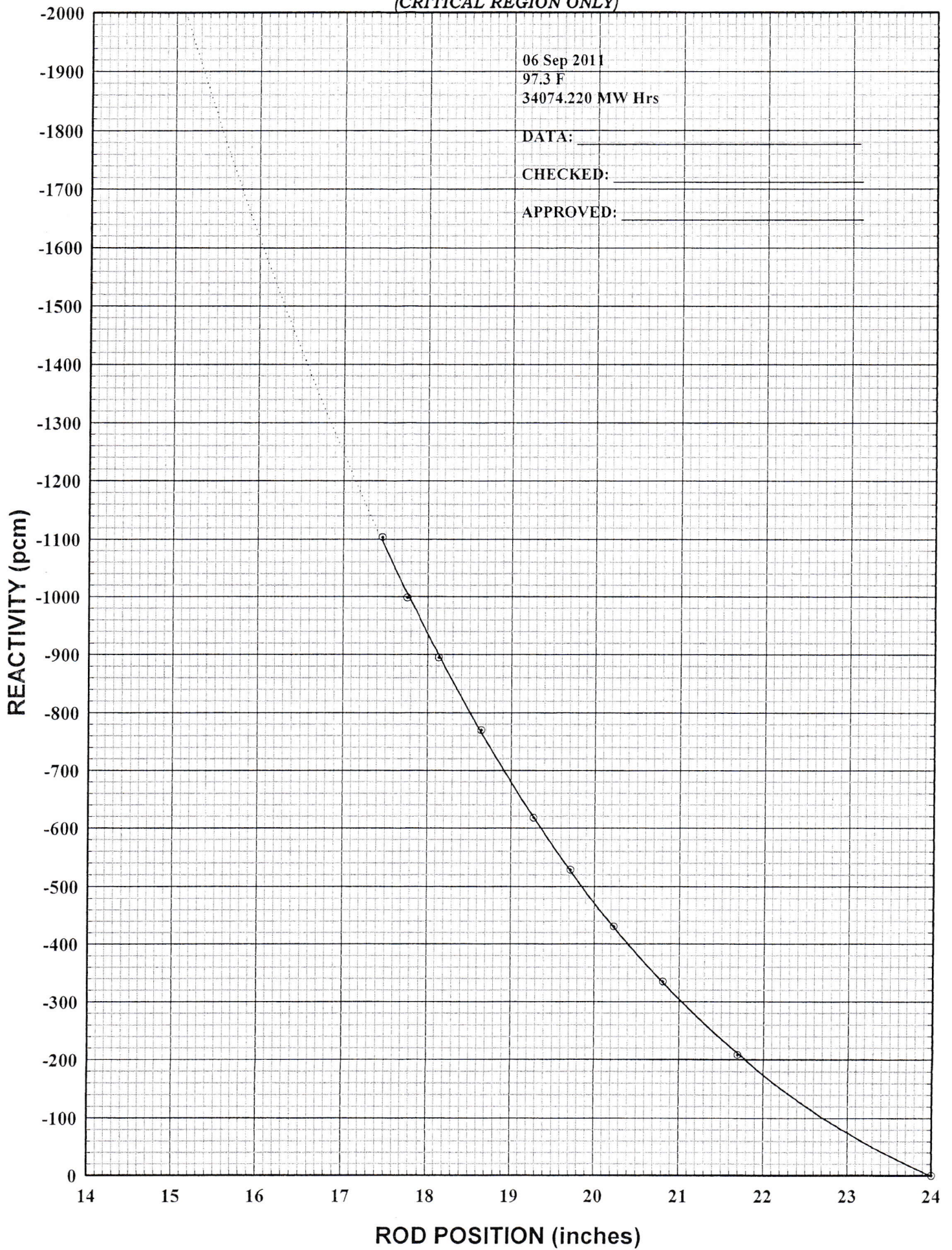
3. Given (ρ_{excess}) above, **calculate the Shutdown Margin (pcm)** if the total Gang Rod Worth is – 8000 pcm fully inserted at 0" Gang Rod Height.

POWER DEFECT LAB:

1. Which temperature related reactivity coefficient(s) are accounted for by the power coefficient of reactivity (i.e. power defect)?
2. **P**oint **O**f **A**dding **H**eat (POAH):
 - a. What is the POAH as quoted in the NE235 theory manual?
 - b. What was the ~ POAH as observed in lab (reference lab data spreadsheet)?
3. What was the total power coefficient we measured in lab at 750 kW?
4. What were the main sources of uncertainty in the measurement of #3 above?
5. Was power defect (pcm/MW) completely linear (R^2 fit parameter = 1.000?) with respect to power?

GANG INTEGRAL ROD WORTH

(CRITICAL REGION ONLY)



XENON REACTIVITY AFTER SHUTDOWN FROM 1 MW OPERATION WITH DURATIONS AS GIVEN

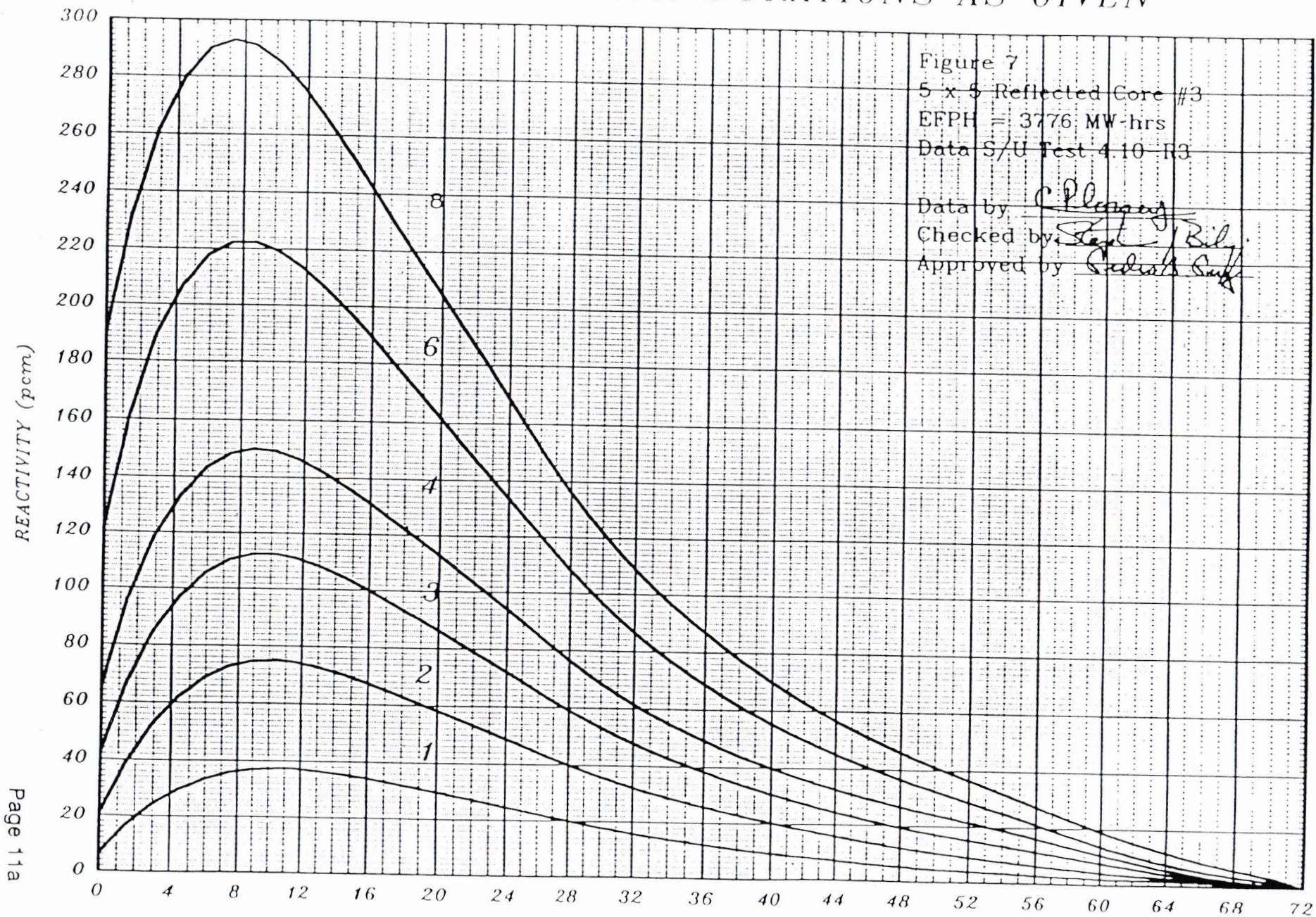


Figure 22

Revision 27
07 September 1993